HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and deployed **agentic AI systems** for industrial operations, including orchestration, tool use, and human-in-the-loop workflows, directly contributing to the development of a next-generation industrial operating system.
- Developed **multimodal data pipelines** integrating **video processing**, **computer vision** (YOLO, OpenCV), and structured operational data from sensors and machines to improve efficiency and decision-making.
- Built **signal-processing and sensor-correlation workflows**, including **FFT feature extraction** and cross-sensor analysis on time-series telemetry, vibration, and thermal data streams.
- Implemented **REST APIs** using **Java Spring Boot** for cross-service data exchange between enterprise systems and the industrial platform, ensuring data normalization and consistent schema mapping.
- Containerized backend services with **Docker** and deployed them to a **Kubernetes** cluster (EKS), managing configuration and resource allocation for predictable rollouts.
- Engineered **Python data pipelines** on AWS Lambda + S3 to ingest and process industrial data, handling approximately **3,500 events/sec** at peak hours.
- Developed and shipped **FastAPI inference endpoints** for ML models, achieving a **p95 latency of 180ms** and implementing input validation for production AI systems.

SKILLS

AI/ML – Agentic AI, Multimodal Machine Learning, Computer Vision, YOLO, OpenCV, Signal Processing, FFT, Predictive Maintenance, Operations Planning, AI Systems, Model Serving, Inference

Data Pipelines & Integration – Data Ingestion, Normalization, Temporal Alignment, Data Quality, ERP Integration, CMMS Integration, WMS Integration, Industrial Historians, PLC/OPC-UA, Knowledge Graphs, Industrial Ontologies

Programming Languages – Python, Java, TypeScript, SQL, R

Backend Frameworks & Technologies – Spring Boot, FastAPI, Flask, NestJS, Node.js, REST APIs

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, Jenkins, Git, CI/CD

Databases – PostgreSQL, MySQL, Oracle, Vector Databases

PROJECTS

Industrial Intelligence Platform Core AI

- Designed and built core **AI capabilities** for an industrial intelligence platform, connecting advanced models with operational data and enterprise systems.
- Developed **agent runtimes and orchestration frameworks** supporting specialized agents for vision quality and predictive maintenance, integrating with **OpenCV** and **YOLO**.
- Implemented **context assembly systems** using industrial ontologies and memory, including tool dispatch, approval gates, and tracing for complex industrial workflows.

Real-Time Sensor Data Analysis Pipeline

- Engineered a **Python pipeline** to process and analyze real-time industrial sensor data, including **RTSP video streams** and time-series telemetry.
- Implemented **signal processing techniques** such as FFT for feature extraction and conducted cross-sensor analysis, reducing noise by approximately **30%**.
- Integrated the pipeline with a **PostgreSQL** database for storing processed data and facilitating downstream analysis for operational decision-making.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)